# ZOO502 FINAL TERM CURRENT PAPERS

## (MIDTERM CURRENT MCQS INCLUDED)

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## FINAL TERM CURRENT MCQS OF ZOO502

1. Non diffusible ___produce Donnan effect SOLUTE

2. _____ channels are involved in producing action potential. Na+ and K+

3. Which statements not related to voltage gated Na channel._____ options likh k laye

4. AChE abundantly present in SNYPTIC CLEFT

5. Which ___monitors the position of muscles and joints. PROPRIOCEPTORS

6. Sensory sensilla are organ of ___in insects. TASTE

7. Merkel Disks are associated with the reception of VIBRATION

8. Human ear can detect sound 20 TO 20000 HRTZ

9. interstial tissues constitute___of adult testes mass 20%

10. Adenohypophysis is the site for production of ___ options likh k laye

11. Velocity of impulse conducting by arterial fiber is 0.3 m/s

12. Equation indicate blood flow **V = Q/A**

13. Relative viscosity of plasma 1.8

14. Fenestrated capillaries are found in endocrine glands

15. Which is local vasoconstrictor Angiotensin II

16. ___ is formed in lungs, acts as local vasoconstrictor Angiotensin II

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17. Amount of air M/V that still remains in lungs when they fully collapse 30 to 120mL

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18. Respiratory system of ___ is composed of 7 or 9 air sacs Birds

19. Central chemoreceptors are located in ____ Medulla of mammals and Air breathing vertebrates

20. which are stenohaline animals Marine spider, crab

21. A camel can lose ___ of water and still survive 40%

22. ___ is involve in maintaining pH of urine TUBULAR SECRATION

23. GFR averages an astounding ___ per minute 125 mL

24. It requires only __ of water to excrete 1g of urea. 0.05 L

26. Esophagus is included in ___ division of mammalian digestive tract. Four (4)

27. Human GIT secrets__ exocrine fluid daily. Bile and Bile salts

28. Goblet cells synthesis and secretes Mucus

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29. ___ is largest water soluble essential nutrient Vitamin B12

30. Brown fat contains large amount of Mitochondria

31. Which is Neuroethological behavior Study behavior in laboratory

33. Agonistic behavior involve___b/w animals Competition

34. Social behaviors are based on Competition

35. Respiratory epithelium has only a thickness of 0.5 to 15 um

36. High pressure is Systolic BP

37. Atrial natriuretic peptide (ANP) hormone secreted by Stretch-Sensitive Secretory Cells

38. Blood with hemoglobin carry ___ Oxygen per 100ml. 20 ml

39. Inspiratory volume is 1900 ml

40. DCT secrete _____ NH4+ ions, K+ ions, & H+ ions.

41. Teloest fish is _____ Ammonotelic Animals



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42. Esophagus is part of Foregut

43. Deudenum is _____cm long 25cm

44. Troponin is sensitive to Ca+

45. Electric impulse is observed in Cardiac muscle

46. Proteases are (trypsin , chymotrpsin , pepsin)

47. Facilitated diffusion occurs with channel proteins

48. Sweating occur in UCT (Upper Critical Temperature)

49. Thickness of muscle fibers is ____

50. 1 joule of energy utilized in work, 3Joule energy is degraded

51. Study behavior in laboratory Neuroethological

52. A honey bee finds a food source close to the 50meter

53. One female has an exclusive relationship with two or more males' poly androus

54. Social behavior (Cooperation , competition, conflict)

55. Agonistic behavior (threat display, attach and fight)

## MIDTERM PAST CURRENT MCQS OF ZOO502

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1 A slowly occurring evolutionary process Adaptation

2 Changes are induced experimentally in the laboratory Acclimation

3 Neurotransmitters induce large increase in permeability to ions by acting on integral membrane proteins in___ nerve and muscle cell

4 Diffusion through the membrane down the conc. gradient with the help of transport proteins facilitated diffusion

5 Donnan equilibrium is characterized by a reciprocal distribution of the anions and cations

6 The conc. of K+ is maintained 10-30 times more in the cytosol than extracellular fluid



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7. carrier protein Na+/K+ ATPase
8. A depolarization in the range of -55 to -30 mV is the threshold that triggers an AP.
9. The permeability of Na+ channel for K+ ions 0.09
10. Giant axons of arthropods and molluscs: 30 m/sec
11. Which of the following is not a pharmacological application Anemia
12. Calcium ions are not delayed rectifiers
13. Example of mechanical stimuli is gravity
14. Which one is not G protein receptor for taste sour and salty
15. Merkel's Disks are associated with the reception of vibration
16. Meissner's Corpuscles receptors are for touch
17. Each hair cell has 20-300 nonmotilestereocilia.
18. Human ear can detect sound frequencies lying between 20 to 20000 hertz.
19. Hair cells have a Vrest of -60 mV
20. Cold Receptors are 3.5 times more common in skin than heat receptors
21. Rods are more sensitive to light
22. Sensory layer of eye with photoreceptor cells Rods and cons is Retina
23. Visible spectrum for human eyes lies between the wavelengths of 400-740 nm
24. A person with loss of red cones is called a protanope
25. Effect on neighboring cells paracrine secretion
26. Adenohypophysis has pituitary lobe anterior
27. ACTH is a peptide hormone comprising single chain of 39 amino acids
28. Hypothyroidisim causes Cretinism and Goiter
29. Regulate Ca2+ and phosphate ions in blood Parathormone PTH
30. Adrenal cortex produces two major types of steroid hormones: • Mineralocorticoids •Glucocorticoids
31. CatecholaminesConversion of tyrosine to dopa and dopamine occurs in the cytosol
32. Delta Cells constitute 10% of islets
33. interstitial cell constitutes of 20% mass
34. placental estrogens are 30 times more than the normal levels.
35. Thyroglobulin is stored within the thyroid gland in large follicles
36. Diameter of membrane channel is less than 1.0nm
37. channel involve producing action potential Na+ and K+
38. Action potential is inflow of Na+ depolarization
39. Hallucination Endorphins

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4. The receptor molecules are typically proteins
4. sensory sensilla are organ of taste
4. Prolactin regulates fat metabolism and reproduction in Birds
4. Example of peptide hormone in reproduction hormones Oxytocin
4. Rod photoreceptors are sensitive to low intensity dimmer light
4. Concentration of K+ is 10-20 times higher inside the cell than outside
4. The correct statement is Symporters are coupled transporters that transfer two solutes in the same direction
4. Most neurons have an RMP of -70mV, when no impulse is being conducted
4. Radioactively labeled TTX is used to estimate density of Na+ channels
4. Voltage-Gated Na+ Channels are Fast Acting channels
5. In many invertebrate species, APs can last for 10-100 milliseconds
5. Delayed Rectifiers are Voltage gated K+ channels
5. Cytosolic conc. of Ca2+ remains below $10^{-6}$ M
5. Major ions responsible for Vrest are K+, Na+ &Cl-
5. Which of the following is not correct about ion channel protein Have Hydrophilic passageway that facilitate quick flow of water molecules and ion in dissolved form.
5. Thickness of cell membrane is 5nm
5. Which is not associated with ATPase Ca Activated
5. Which is not Fast inhibitory transmitters Acetylcholine
5. AChE is abundantly present in the synaptic cleft
5. Mescaline that induces hallucinations
6. The genes for red and green-cone pigments are closely linked on the X chromosome
6. Example of steroid hormone is Aldosterone
6. substance that is released via a duct through the gland to the external or internal epithelial surfaces Exocrine secretion
6. Inhibits release of Growth Hormone (GH) Somatostatin
6. Amine hormone (Thyroid & Adrenal Modulary hormone) are synthesized by the action of Enzyme on amino acid Tyrosine
6. Acts as enteric neurotransmitter Substance P
6. In many organisms, ACTH also plays role in circadian rhythms
6. Pituitary Gland release __ hormones 9
6. Which is not part of a cell body Myelin Sheath

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6. Adenohypophysis It contains ___five___ types of glandular cells that synthesize and secrete six hormones
7. A color-blind person who lacks ___green_____ cones is called a deuteranope
7. Example of Apocrine secretions ____Mammary gland__
7. Salivary glands produce saliva that is delivered to the oral cavity through __submandibular and parotid ducts
7. Oxytocin stimulates motility of the oviduct in ___Birds___
7. Which of the following hormone is released from anterior lobe of pituitary gland__ luteinizing hormone LH
7. Parathormone over secretion leads to ____Softness of bones__
7. Type of secretion in which entire cell raptures and breakup to release its content__Holocrinesecration__
7. Human retina contains ___6 million___ cones
7. Which one is not related to diffusion option_____ low to high concentration
7. melatonin belong to_____ pineal Gland
8. pineal gland found dorsal surface of.....in vertebrate _____forebrain
8. target cell of hormones usually have ....receptors _____2000 to 100,000
8. Thick filament are composed of ...._____ myosin.
8. Each molecule of calsequestrin protein can find around.....calcium_____ 50
8. human s-A node is ...thick_____ 1 millimeter thick
8. Velocity of impulse conduction through most atria fiber is....._____ 0.3 m/s.
8. Average cardiac output for resting adult man-_____ 5L/mint
8. Myocytes are ____ muscles _____Cardiac
8. The knob of dendrite has ____olfactory cilia_____ 4 to 25
8. Synaptic cleft is ___ nm wide _____20
9. Pituitary gland is __ dm _____1cm
9. Rhythmical discharge of S-A node is: ____/mint _____70 to 80
9. An adaptive change within an individual animal. Happens due to chronic exposure to new, naturally occurring environmental conditions____ Acclimatization.
9. Antidiuretic hormone can increase water permeability of renal collecting duct in mammals up to _____10 times
9. _____ are delayed rectifiers _____ Voltage – gated $K^+$ channels

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9. Which one of the following is not the feature of the Voltage –gated $Ca^{+}$ channel _____ They are known as delayed rectifier

9. Which of the following statement is correct about action potential ____ inflow of sodium ion (depolarization) Download More Files from VUAnswer.com

9. Which of the following monitor the position of muscles and joints____ Proprioceptors

9. Taste receptors in terrestrial vertebrates are not located in/on _____ Larynx

9. Which is not the part of organ of Corti_____ Reissner's men

100. Near the end of pregnancy, placental estrogens are 30 times more than the normal levels. _____ 30 times

101. Myofibers are ___ in diameter, and up to many centimeters in length____ 5 to 100 µm

102. Troponin has high affinity for_____ Ca2+.

103. When the force exerted by the muscle contraction is equal to the opposing external force than ___ contraction __isometric

104. Atrial contraction provides only __volume of total mammalian ventricular output____ 30%

## Subjective

## 2 Marks:

1. **What is blubber and its role /2**

**Blubber** is a thick layer of fat, also called adipose tissue, directly under the skin of all marine mammals.

**Role:** As fatty tissues are little metabolically active & have little blood vascularization so it is a good insulator b/c of low thermal conductivity and prevent heat loss through body surface.

2. **Define peristalasis /2**

Waves of contraction and relaxation in the smooth muscles present in the walls of alimentary canal. The alimentary canal starts from oral cavity and end at distal end of rectum. Esophagus conducts the swallowed bolus by peristaltic movement.

3. **Q. Write down the basic types of alimentary system? 2.**

**Types of Alimentary Systems:**

Two basic plans of alimentary systems in animals:

• Sac-like gastro vascular cavity • Tubular alimentary canal or gut

4. **Q. How alveolar collapse is prevented? 2**

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**Prevention of Alveolar Collapse:** Alveolar collapse normally does not occur for two reasons

1. Surrounding tissue prevents over-expansion of alveoli.
2. Presence of pulmonary surfactants.

**5. Q. Differentiate between laminar and turbulent flow? 2**

**Laminar Flow:**

Flow is streamlined and continuous. The velocity of fluid is constant at every point. Flow occurs in layers at different velocities. Flow is zero at the wall and maximal at the center, A pressure difference supplies the force required to slide adjacent layers to past each other.

**Turbulent Flow:**

Direction of fluid movement is not aligned. It requires higher energy to move blood through Vessels. Turbulence also happens if smoothness is reduced by any obstruction e.g. Buildup of fatty deposits on arterial walls.

**6. Q. Name mechanoreceptor fiber in the arterial wall? 2**

Ans: **Atrial Receptors:** Atrial walls have mechano-receptive afferent fibers including:

- Myelinated A-type afferent fibers
- Myelinated B-type afferent fibers
- Unmyelinated C-type afferent fibers
- Stretch-sensitive secretory cells

**7. Fahraeu- lindqvist effect? 2**

**Fahraeus-Lindqvist Effect**: Phenomenon of plasma skimming & resultant increased blood flow velocity is called Fahraeus Lindqvist effect. It reduces energy required to drive blood through microcirculation

**8. Types of vascular resistance /2**

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**Vascular Resistance** "The resistance that must be overcome to push blood through the circulatory system and create flow" **Types** • Systemic vascular resistance (SVR) • Pulmonary vascular resistance (PVR)

**3 Marks:**

**9. Objectives of ethology /3**

**Ethology:**

"Scientific study of animal behavior in natural environment" .

Objectives of Ethology: Involves understanding of

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- Stimuli that elicit behavior
- Physiological mechanisms mediating response
- Animal's experiences that influence behavior
- Role of behavior in survival & reproduction
- Evolutionary history of behavior

**10. Four accessory glands of elementary canal? 3**

Accessory Glands: Secrete digestive juices through ducts. The accessory glands of the mammalian digestive system are:

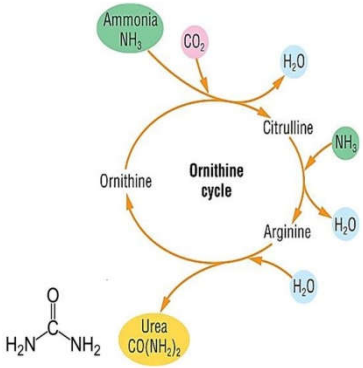
- Three pairs of salivary glands
- Gastric glands
- Pancreas
- Liver
- Gallbladder

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**11. Briefly describe ornithine urea cycle /3**

**Ornithine-Urea Cycle**

- Used by all vertebrates except teleost fishes
- Synthesize urea primarily in the liver
- Two ammonia and one CO2 molecule are added to ornithine to form arginine
- Enzyme arginase cleaves arginine to remove a molecule of urea and regenerate ornithine




**12. Factors affects systemic vascular resistance /3**

**Factors affecting SVR**

- Elasticity of vessel wall
- Diameter of vessel
- Decreasing vessel diameter (vasoconstriction) increases SVR
- Increasing vessel diameter (vasodilation) decreases SVR

**13. 3 uses of crop in various animals.(3)**

**Crop**

• A sac-like expanded section in esophagus of some animals. Used to store food without digesting.

Crop plays various roles in different animals. For example

- Food storage without digestion

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- Fermenting the food
- Prepare food for nestlings (pigeon's milk)

**14. Q. Name 3 vasodilators in the body? 3**

1. Histamine released from connective tissue and white blood cells in injured tissues causes vasodilatio
2. Plasma kinins (e.g. bradykinin) are potent vasodilators in damaged tissues
3. Nitric oxide (NO) serves as a major inducer of vasodilation in cardiovascular system also causes relaxation of vascular smooth muscle and acts as vasodilator
4. Lactic acid generated by tissue cells
5. Rising conc. of K+ or H+ ions in interstitial fluid
6. Prostacyclin causes vasodilation and acts as an anticoagulant.

**15. Write the name of local vasoconstrictor?**

Ans:

1. Prosta gland and thromboxanes released by activated platelets and white blood cells.
2. Endotheline released by damaged endothelial cells.

**16. Write down the composition of the Bile? 3**

Ans: **Bile:** Vertebrate liver secretes bile. Bile doesn't contain any digestive enzymes It contains many salts Bile salts are essential for the digestion of fats

**Composition of Bile:** Bile is a weakly basic mixture of: Water, Cholesterol, Lecithin, Inorganic salts, Organic salts, & Bile pigments. Organic salts of bile are manufactured in liver from cholesterol and amino acids complexed with sodium Bile pigments are derived from biliverdin and bilirubin—products of breakdown of hemoglobin coming from ruptured red blood cells.

**5 Marks:**

**17. Q. What is Rete Mirabile? Mention its function in the Tuna fish? 5**

Ans: **Rete Mirabile** A highly evolved system of countercurrent heat exchange. Found widely in homeothermic and
heterothermic vertebrates. The artery carrying warm blood toward extremity is completely encased in a circlet of veins carrying back cold blood.

**Function in the Tuna fish:**

Forms an extensive heat exchange surface b/w inflowing and outflowing blood. Large number of rete mirabile in tuna (fish) are used to regulate temperature of brain, muscles, and eyes. Swim

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bladders of some fishes (e.g. eels) are associated with rete mirabile and functioning as CO2 exchangers.

**18. Note on physiological adaptation of ectothermic in hot environment /5**

**Physiological Adaptations**

- Ectotherms have high heat conductance
- Radiate heat rapidly—by moving to less warmer places
- Panting: Above 40oC many reptiles start panting—like birds and mammals
- Increases heat loss through respiratory evaporation, causing cooling.
- Regulating heart rate and flow of blood to surface tissues—to eliminate heat (e.g. marine iguanas)
- Gila monster (Helodermasuspectum) evaporates water from its cloaca to cool—a process similar to sweating in mammals

**19. Write a note on types of stomach in vertebrates /5**

Types of Stomach in Vertebrates: Based on number of chambers, stomachs are classified as:

- Monogastric stomachs
- Digastric stomachs

**Monogastric Stomachs**

- Consist of a single strong muscular tube or sac
- e.g. stomachs of carnivorous and omnivorous vertebrates

**Digastric Stomachs**

- Multichambered
- Found in ruminant mammals of order Artiodactyla
- Examples: deer, giraffe, sheep, cattle, camel, llama

**Digastric Stomachs**

- These stomachs have four chambers, with two divisions:
- First division consists of fermentation chambers rumen and reticulum
- Second division comprises true stomach (omasum and abomasum)

**20. What is hypoxia? Responses of aquatic animals in it /5**

Hypoxia • “A condition in which the body or a region of the body is deprived of adequate oxygen supply”

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**Hypoxia in Aquatic Environment**

• O2 levels in aquatic environment vary more frequently

• Aquatic animals are faced with frequent and rapid changes in oxygen levels. Many aquatic animals can withstand very long periods of hypoxia

**Responses of Aquatic Animals to Hypoxia:**

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- Many aquatic animals utilize anaerobic metabolic pathways to survive hypoxic periods
- Many animals adjust respiratory & cardio-vascular systems to maintain O2 absorption from hypoxic H2O
- Many fishes increase gill ventilation rate in hypoxic conditions
- Ram ventilators (e.g. tuna) widely open mouth to increase water flow over the gills

**10 Marks:**

**21. What are pheromones? Also describe their Role? /10**

**Pheromones:**

• Chemical messengers released by many animals in environment to communicate. Spread through air.

**Characteristics:**

• Affect the behavior of individuals of a species

• Effective at remarkably low concentrations

• Have particular odors

• Detected by olfactory receptors of receiving animal

**Roles of Pheromones:**

**1. Recognize Members of Species:**

Unicellular organisms recognize members of their own species. Many insects identify members of their colony.

**2. Reproductive Roles:**

Pheromones attract members of opposite sex for breeding.

Induce courtship behaviors in members of a species.

Induce spawning behavior in aquatic animals with external fertilization.

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**3. Territorial Behaviors:**

Many male mammals mark their territories with pheromones.

It warns other males of their occupied area.

**4. Insect's Social Order:**

In a honeybee colony, pheromones produced by queen maintain hive's complex social order.

**5. Alarm Calls:**

Members of some species produce pheromones on getting injured.

It warns other members by the presence of danger in area.

**6. Repel Predators:**

Pheromones produce by the pray may repel predators.

Example: The skunks produce foul-smelling musk that makes skunks unpalatable or distasteful to their enemies.

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